FLEET TEAM DIGEST

737MAX-FTD-32-17005

Issue Title : Reuse of Axle Nut Retaining Fasteners

Airplane Model	ATA	Minor Model(s)
737MAX	3240-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
757, 747-400, 767, 747, 777, 737NG	9E991

Originated Date	Last Revised Date	Created On
05/02/2017	05/02/2017	04/11/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

Boeing has received a few reports of loose or lost axle nut retaining bolts and/or locknuts on various Boeing airplane models over the past 30+ years. In some of these cases, excessive reuse of the lockbolts and locknuts is believed to have been the root cause. Omission of the lock bolt/nut during axle nut installation is also a possible cause.

At least one wheel damage event and one wheel loss event on 747-series airplanes occurred due to loss of the axle nut retaining bolts. In both of these events, it is believed that the retaining bolt locknut usage was not controlled, with the locknuts being reused a large number of times, resulting in loss of self-locking capability.

Background

Typically, the most severe airplane-level effect of a loosened axle nut retaining bolt is damage to the inside of the hubcap (caused by contact between the loosened retaining bolt head and hubcap). However, if both retaining bolts back out, the axle nut may loosen, resulting in a decrease of wheel bearing preload. This preload decrease can cause wheel bearing damage/failure and loss of the wheel assembly.

Status

Not applicable.

Interim Action

Not applicable.

Final Action

Not applicable.

Operator Action

Operators are encouraged to control the number of times axle nut retaining bolt locknuts are reused. This action will help minimize the potential for loose or missing locknuts and lockbolts and the potential for a wheel bearing failure and/or wheel loss.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

FLEET TEAM DIGEST

While locknuts are typically qualified using installation and breakaway torque values from fifteen installation cycles, operators may wish to adopt a policy to regularly replace the locknuts and lockbolts at some convenient interval specified by the operator based on their operation and service experience.

If the axle nut retaining lockbolts and locknuts are reused, it is important to ensure that the locknuts retain their self-locking capability and the lockbolts are in serviceable condition. The self-locking torque values are provided in AMM 20-50-11 for the 737 airplanes.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	32-45-21/401		
Aircraft Maintenance Manual (AMM)	32-45-11/401		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- In Service Occurrences : null

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991